WORLD'S STRUCTURE

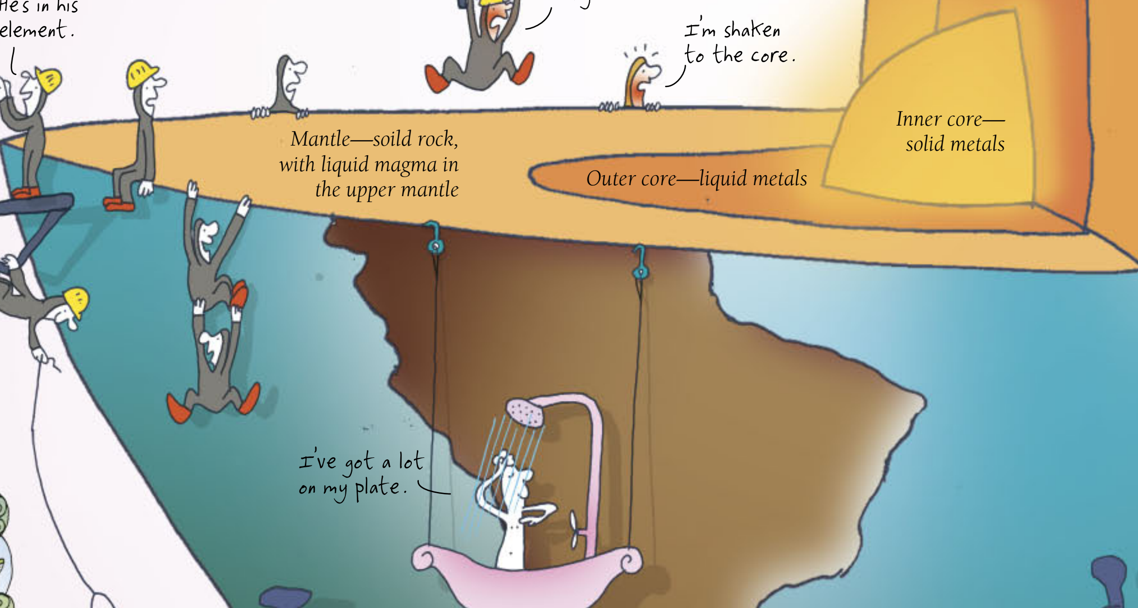
Our Earth is a complex ball of different chemical elements combined together to form rocks and minerals that are constantly on the move. The tectonic plates that, pieced together, make up Earth's shell are constantly shifting and breaking up the landscape. As continents drift and jostle across Earth's surface, massive mountain ranges are thrown up, volcanoes erupt, and earthquakes shake the ground beneath our feet.

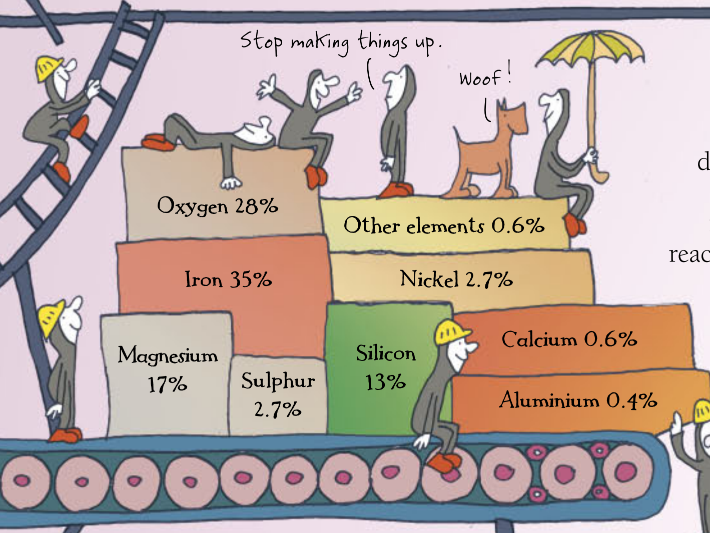
Earth's layers

Earth is made up of different layers, with a solid metal inner core, made hard by immense pressure, and an outer core of molten metals. Wrapped around the core is a layer of solid rock, which turns molten as it edges toward Earth's crust—the rigid layer of Earth's shell.

Inside Earth

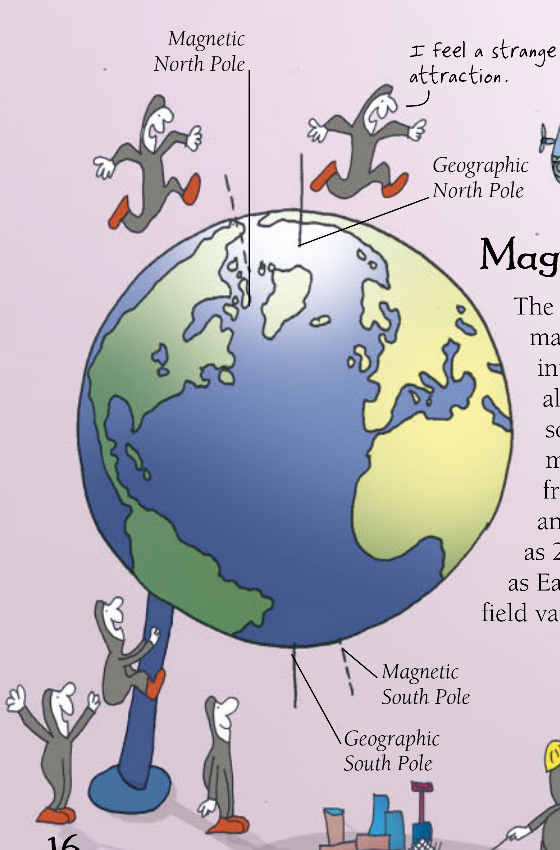
Dig just below the surface of the Earth and the temperature drops slightly. However, from there on down the temperature rises to be fantastically hot, so that by the time you reach Earth's inner core, the temperature has soared to a scorching 6,700°F (3,700°C).





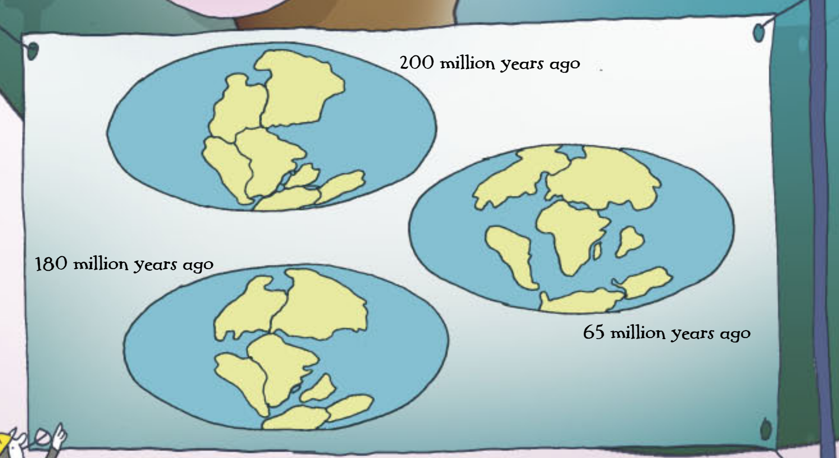
Chemical makeup

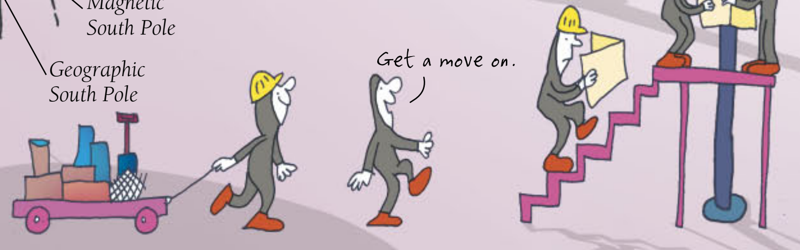
More than 80 separate elements make up Earth. The largest component is iron, which is thought to be found largely in the core. Oxygen, magnesium, and silicon are also important elements in Earth's structure, and occur in large quantities.



Magnetic Earth

The dense core of iron that makes up Earth's core turns it into a giant magnet which, like all magnets, has a north and south pole. These two magnetic poles are different from the geographical poles, and move around as much as 25 miles (40km) a year as Earth's magnetic field varies.





On the move

The world's continents sit on top of moving tectonic plates that float on Earth's upper mantle. These plates were once joined together, but gradually broke up and drifted apart to form the continents we know today.